



Choupo

The Open Source Chemical Process Simulator

User Guide

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Preface — read this first

Engineers used to text-driven, file-first tools will feel at home. Newcomers will feel curious, then liberated. This is an *educational* process simulator: every Newton iteration, every K -value, every Wegstein step is visible both in the source and at runtime. If something surprises you, the answer is always one **grep** away.

This guide walks you from a cold install to running, modifying, and building your own cases. A dozen-odd short chapters; each opens with a *What you'll learn* note and closes with a hands-on cue. Boxes mark crystal-clear takeaways, common pitfalls, and small experiments you should run as you read.

1 Quick start

What you'll learn. How to build the simulator once, set up your shell environment, and run your first tutorial in under five minutes.

Build once, source the shell environment once, then run any tutorial:

```
cd /path/to/Choupo
make all                      # release build
source etc/bashrc            # adds bin/ + project root to PATH

listCases                    # available tutorials
runCase tutorials/steady/flash/flash01_benzene_toluene # run one
less tutorials/steady/flash/flash01_benzene_toluene/log.choupoSolve
```

Tutorials live under `tutorials/<category>/<name>/`, where the category is one of `steady/` (`choupoSolve`), `batch/` (`choupoBatch`), `ctrl/` (`choupoCtrl`), `props/` (`choupoProps`), or `plant/` (multi-sector cases). `runCase` reads each case's `controlDict.application` and dispatches to the right binary automatically — you never pick the executable by hand. After your first run a symlink (e.g. `./choupoSolve`) points to the matching `build/linux64Gcc/` binary.

1.1 Permanent shell setup

Append this line to `~/.bashrc` so every new terminal has the environment ready:

```
source /path/to/Choupo/etc/bashrc
```

A fresh terminal then greets you with:

```
Choupo environment loaded.
CHOUPO_HOME = /path/to/Choupo
Binaries      : choupoSolve, choupoBatch, choupoCtrl, choupoProps
Commands available: runCase, runApplication,
                  listCases, cleanCase, cdOPE, cdTut
```

1.2 Build modes

```
make all          # default --- release, -O2
make MODE=debug all # -O0 -g, parallel tree under build/linux64GccDebug
make print        # show resolved variables
make clean        # remove current mode
make distclean    # remove the whole build/ tree
```

Both modes coexist; the `choupoSolve` symlink points to the *last* binary built.

Key idea. You never recompile to switch cases. Build once, source once, then all you do is edit text dictionaries and re-run.

Try it now. Run `runCase tutorials/steady/flash/flash01_benzene_toluene` and then `less tutorials/steady/flash/flash01_benzene_toluene/log.choupoSolve`. Notice the per-iteration trace under “Rachford-Rice”: that is the simulator showing its work.

2 The four shell helpers

What you'll learn. Five short commands cover every interaction you'll have with the simulator from the shell.

After sourcing `etc/bashrc` you have:

Command	Purpose
<code>choupoSolve <case></code>	Run a case; write everything to stdout.
<code>runCase [<case>]</code>	Run + redirect to <code>log.choupoSolve</code> in the case dir. Defaults to current dir.
<code>runApplication choupoSolve <args></code>	redirect to <code>log.<appName></code> in cwd.
<code>listCases</code>	Tabulate all tutorials with one-line descriptions.
<code>cleanCase [<case>]</code>	Remove <code>log.*</code> and <code>*.log</code> from a case (never touches source).
<code>cdOPE</code>	<code>cd</code> to the project root.
<code>cdTut</code>	<code>cd</code> to <code>tutorials/</code> .

Two equivalent workflows:

```
# From anywhere (use $CHOUPPO_HOME):
runCase $CHOUPPO_HOME/tutorials/steady/flash/flash01_benzene_toluene

# from inside the case dir:
cd tutorials/steady/flash/flash01_benzene_toluene
runCase                                # writes log.choupoSolve here
```

3 Anatomy of a case directory

What you'll learn. Every case is a folder of plain-text dicts. Two are required; three are optional. Nothing else.

```
caseName/
|-- system/
|   |-- controlDict          required  --- verbosity, [future] time control
|   |-- flowsheetDict        required  --- topology: streams + units + connections
|   |-- solverDict           optional  --- per-unit-op solver options (overrides)
|   |-- outerDict            optional  --- sensitivity / optimisation / param-est
|   '-- postDict             optional  --- post-processing chain (sizing, costing)
|-- constant/
|   |-- thermoPackage         required  --- components + gamma-phi models / phases
|   '-- reactions             optional  --- named-reaction library (CSTR / PFR)
'-- log.choupoSolve          produced  --- written by runCase
```

Key idea. A standalone unit operation is just a `flowsheetDict` of length 1. There is no special “single-unit” mode — one consistent case format for everything.

3.1 controlDict

The simplest valid file:

```
verbosity 3; // 0 silent / 1 errors / 2 warnings / 3 info / 4 debug
```

For pedagogical use you want `verbosity 3` — that prints every Newton iteration, every Wegstein step, every K -value evaluation. Set lower only when sweeping the simulator thousands of times.

3.2 flowsheetDict

Streams + units + (optional) tear-stream declaration for recycles.

```
streams
{
  feed
  {
    F      100.0 kmol/h; // industrial unit; parser
    T      370.0 K;     // converts to SI canonical
    P      1.0 bar;     // (kmol/s, K, Pa) on the fly
    composition { benzene 0.40; toluene 0.60; }
  }
}

units
(
  {
    name      flash01;
    type      isothermalFlash;
    in        feed;
    outputs   (liquid vapor); // names assigned to producedStreams[0..k]

    operation { T 370.0 K; P 1.0 bar; }
  }
);
```

Units on every scalar. Since, any scalar in a dict may carry a unit suffix. The parser converts to canonical SI internally (Pa, kmol/s, K, m, m², m³, J, kg) and tags the entry with its physical dimensions, so a typo (P 1 kg;) is caught with a crystal-clear error before the solver runs. See §3.3 for the three accepted forms and the `controlDict.units {...}` block.

Composition basis. Five ways to declare a stream's flow and composition — pick exactly one:

```
// (a) F + molar fractions
F          100.0 kmol/h;
molarComposition { benzene 0.50; toluene 0.50; }

// (b) F + mass fractions
F          5000 kg/h;
massComposition { benzene 0.40; toluene 0.60; }

// (c) legacy alias of (a)
F          100.0 kmol/h;
composition { benzene 0.50; toluene 0.50; }

// (d) per-species molar flows -- F and x_i derived
molarFlows
{
    benzene  40.0 kmol/h;
    toluene  60.0 kmol/h;
}

// (e) per-species mass flows -- F and x_i derived
massFlows
{
    benzene  3124.4 kg/h;
    toluene  5528.4 kg/h;
}
```

Forms (d) and (e) are the natural way to describe a feed in process design (“mix 40 kmol/h of A with 60 kmol/h of B”); F total and composition are computed from the sum. Forms (a) / (b) / (c) are the textbook way (total flow + composition).

Internally the simulator always stores mole fractions, so mass-basis inputs (b, e) are converted at parse time via $x_i = (w_i/MW_i) / \sum_j w_j/MW_j$.

Sum check. A composition block is always renormalised to $\sum_i x_i = 1$, but the parser also checks the user-supplied sum and either passes silently or shouts, depending on how far it drifts from 1. Four regimes:

Input	Behaviour	Why
$\sum_i = 1.0$ exact	silent — values used as data	textbook input
$ \sum_i - 1 \leq 10^{-3}$	silent — normalised	rounding drift
$ \sum_i - 1 > 10^{-3}$	warning + normalised	likely typo
$\sum_i = 0$	fatal error — simulator stops	no feed at all

The warning emitted on the third case names the stream, the block, and the actual sum:

```
WARNING: Stream 'feed': molarComposition sums to 0.9 (should be 1.0);
        values will be normalised
```

Why renormalise instead of rejecting outright? Two pedagogical conveniences:

- Shortcut for equimolar feeds: { A 1; B 1; } is an unambiguous 50/50 without having to type the arithmetic.
- Tolerance for rounding: { A 0.333; B 0.333; C 0.333; } (sum 0.999) should not abort a simulation.

The threshold 10^{-3} is tight enough to catch a typo like { A 0.4; B 0.5; } (forgot to write 0.6) yet loose enough to wave through legitimate rounding.

Flow-based forms are not normalised. When the stream is declared via `molarFlows` or `massFlows` (forms d / e above), the absolute species flows are the source of truth and the parser sums them to find F_{total} . There is no “should sum to 1” check; the only fatal case is $F_{\text{total}} = 0$ (an empty feed).

Sixth form: state keyword + flash on import. The five flavours above all assume you know T and P ahead of time. But many streams in a problem have an *under-specified* thermodynamic state that the simulator should resolve on import. Two examples that come up immediately:

- Heating utility (live steam at 170 kPa): the engineer knows the header pressure; the temperature is whatever T_{sat} corresponds to it; the flow is whatever the consuming unit demands.
- Cooling water (saturated liquid at supply pressure): same logic, on the liquid side of the saturation curve.

For these streams, the user declares a **state** keyword and only the variables that are physically known. The parser completes the rest by inverting the dominant component’s saturation curve:

```
liveSteam
{
  P          170.0 kPa;
  state      saturatedVapour;    // -> T computed from
                                //      Psat-1(170 kPa)

  molarComposition { water 1.0; }
  // no F! The consuming unit op (the evaporator's heating
  // chest) computes the actual steam demand and the simulator
  // writes it back into the stream registry after solve().
  // The Streams panel then shows the real consumption.
}

coolingWater
{
  T          298.15 K;
  state      saturatedLiquid;    // -> P computed from
                                //      Psat(298.15) for water

  molarComposition { water 1.0; }
  F          50.0 kmol/h;        // here F is known a priori
                                //      (plant-side metered flow)
}
```

Supported **state** values:

state	Computes	Sets vf
saturatedVapour	T from P , or P from T	1.0
saturatedLiquid	T from P , or P from T	0.0
subcooledLiquid	— (validates $T < T_{\text{sat}}(P)$)	0.0
superheatedVapour	— (validates $T > T_{\text{sat}}(P)$)	1.0

Restrictions:

- Currently **state** works for *pure-component* streams (one species with $x_i \geq 0.999$). Multi-component saturated states (bubble-point / dew-point of a mixture) are scheduled.
- If you declare *both* T and P alongside a saturated state, the parser checks consistency against $P_{\text{sat}}(T)$ and throws if they disagree by more than 5%.
- Omitting F is only allowed for streams declared as **saturatedVapour** or **saturatedLiquid** — the utility pattern. Subcooled / superheated streams require an explicit F since they are not demand-driven utilities.

Consistency check on input streams (the bigger picture). The **state** keyword is one piece of the *flash-on-import* contract: every input stream becomes a single point in thermodynamic-state space, fully specified, before any unit op runs. Internally the simulator already does this for ALL streams — whether or not the user wrote **state**. Three checks happen at parse time:

1. Composition normalisation (the four regimes documented in the Sum-check subsection).
2. Phase-state consistency (the **state**-keyword paths above). Without an explicit **state**, a stream with T , P , $\mathbf{z}$ given is taken as “whatever the flash regime turns out to be”; multi-component streams left under-specified will go through a Rachford-Rice flash on first use by a unit op.
3. Dimension cross-check on every scalar value (any unit suffix on T , P , F etc. is validated against the expected physical dimensions; see §3.3).

For recycles, declare the cycle-breaking streams and the Wegstein parameters:

```

tearStreams      (recycle );
recycleTol       1e-5;
recycleMaxIter   60;
recycleWegsteinQmin -0.5;
recycleWegsteinQmax 0.0;

```

3.3 Units

Choupo uses canonical SI internally: Pa for pressure, kmol/s for molar flow, K for temperature, m for length, m² for area, m³ for volume, J for energy, kg for mass, kmol for amount. You almost never want to type those raw, so any scalar value in a dict may carry a unit suffix. The parser converts the value to SI on the fly and remembers what dimension it was — a typo in the unit is caught with a clear error before the solver runs.

Three accepted forms.

```

P    1    bar;           // (a) named-unit shortcut
A_w [-1  2  1  0  0] 1.5e-12; // (b) bracket form
                                   //      [M L T Theta N] + SI value
R    0.5;                // (c) raw SI (no declaration)

```

Form (a) is what you will type 95% of the time. Form (b) is the escape hatch for quantities with no short named unit (membrane permeabilities, exotic transport coefficients). Form (c) says “trust me, the number is in canonical SI” — the parser cannot check anything, so a typo here passes silently.

Named units shipped by Choupo. A non-exhaustive overview of the most useful aliases (see `src/core/Units.cpp` for the full table):

Quantity	Canonical SI	Named-unit aliases
pressure	Pa	Pa, kPa, MPa, bar, atm, psi, mmHg, torr
molar flow	kmol/s	kmol/s, kmol/h, mol/s, mol/h
mass flow	kg/s	kg/s, kg/h, g/s, g/h, t/h
volumetric flow	m ³ /s	m3/s, m3/h, L/s, L/h, L/min
temperature	K	K, degC, C, Celsius, degF, F, Fahrenheit
length	m	m, mm, cm, km, in, inch, ft
area	m ²	m2, m^2, cm2, mm2
volume	m ³	m3, m^3, L, mL
time	s	s, sec, min, h, hr, day
mass	kg	kg, g, t, ton, tonne
energy	J	J, kJ, MJ
power	W	W, kW, MW, power
concentration	kmol/m ³	kmol/m3, mol/m3, mol/L, M
density	kg/m ³	kg/m3, g/cm3, g/mL
viscosity	Pa·s	Pa.s, cP
membrane A_w	m/(s·Pa)	m/s/Pa, m/s/bar

What the student sees on a bad unit. Asking for a pressure but typing `P 1 kg`; produces:

```

ERROR: Dictionary 'feed' ():
  parameter 'P' expected dimensions [1 -1 -2 0 0] (Pa)
  but the dict declared [1 0 0 0 0] (kg). Check the unit
  suffix on this entry.

```

The same check guards every parameter that the unit operations read (`T`, `F`, `V_R`, `area`, `length`, `k_film`, `P_permeate`, ...).

Output preferences. By default Choupo prints flows in kmol/h, pressures in bar, temperatures in K, sizes in m / m² / m³ — the chem-eng convention. To override, drop a `units` block in `controlDict`:


```

units
{
    pressure      atm;          // default bar
    flow          kmol/s;       // default kmol/h
    temperature   K;            // default K
    length        m;
    volume        L;
    energy        kJ;
    power         kW;
}

```

The simulator internals stay in SI; only the printed output changes. Mismatched dimensions in the override (e.g. `pressure m`;) are reported and ignored.

Decimal precision. An optional `precision` block in `controlDict` tunes the number of decimal digits per quantity category:

```

precision
{
    pressure      3;           // default 3
    temperature   2;           // default 2
    flow          4;           // shorthand for molarFlow + massFlow
    composition   4;           // mole / mass fractions
}

```

Other accepted keys: `molarFlow`, `massFlow`, `volumetricFlow`, `length`, `area`, `volume`, `time`, `mass`, `amount`, `energy`, `power`, `concentration`, `density`, `velocity`, `viscosity`, `diffusivity`. Omit any key to keep its default. The defaults match the hardcoded formatting so existing tutorials look identical.

3.4 thermoPackage

Two equivalent syntaxes — legacy (VLE only) and explicit (general, supports LL / VLL):

```

// Legacy:
components      (benzene toluene );
activityModel    { model ideal; }
equationOfState { model idealGas; }

// Explicit:
components (water nButanol );
phases
(
    { name aqueous; type liquid; activity { model NRTL; pairs (... ); } }
    { name organic; type liquid; activity { model NRTL; pairs (... ); } }
);

```

3.5 Optional dicts

- `solverDict` — defaults per unit-op type, overridable by each unit. E.g. `isothermalFlash { tolerance 1e-8; outerAccelerator wegstein; }`.
- `outerDict` — see §6.
- `postDict` — see §7.
- `constant/reactions` — named-reaction library. CSTR / PFR reference reactions by name: `reaction ethanol_to_water_test;`.

4 Built-in unit operations

What you'll learn. The unit-op types available (around two dozen, plus aliases such as `flash`, `HX`, `column`, `FUG`, `membraneSW`) and what each operation block must contain. The table below lists the workhorses; `UnitOperation::availableTypes()` prints the full registry at runtime.

type	What it does	Required keys
<code>isothermalFlash</code> (<i>alias</i> <code>flash</code>)	Multi-component V–L or L–L flash	T, P, phaseSet VL/LL
<code>adiabaticFlash</code>	Adiabatic two-phase flash; solves T_{out}	P
<code>bubbleT</code>	Bubble temperature at given $\mathbf{x}$, P	P
<code>dewT</code>	Dew temperature at given $\mathbf{y}$, P	P
<code>cstr</code>	Single-reaction liquid CSTR	V_R, reaction <name>
<code>pfr</code>	Plug-flow reactor (RK4 over volume)	V_R, nSteps, reaction <name>
<code>heatExchanger</code> (<i>alias</i> <code>HX</code>)	Liquid heater / cooler	one of Tout or Q
<code>mixer</code>	$N \rightarrow 1$ mixer with energy balance	uses inputs (a b c)
<code>splitter</code>	$1 \rightarrow N$ tee with composition split	fractions (f1 f2 ...)
<code>distillationColumn</code> (<i>alias</i> <code>column</code>)	Multi-stage column (Wang-Henke or simultaneous MESH)	nStages, feedStage, refluxRatio, distillateRate, feedQuality, P
<code>shortcutColumn</code> (<i>alias</i> <code>FUG</code>)	Fenske-Underwood-Gilliland shortcut column	lightKey, heavyKey, refluxRatio, P
<code>absorber / stripper</code>	Multi-stage gas absorber / stripper	nStages, P
<code>gibbsReactor</code>	Equilibrium reactor by Gibbs-energy minimisation	T or adiabatic; reactions (...)
<code>compressor / turbine / pump</code>	Rotating equipment; spec $W_{\text{shaft}} + \eta$ (P_{out} is a result)	W_{shaft} , η
<code>electricLoad</code>	Sink for shaft work via an energy wire	uses energyInputs
<code>valve</code>	Isenthalpic throttle to a target pressure	P_{out}
<code>evaporator</code>	Single-effect evaporator with heating duty	P (+ heat)
<code>sprayDryer / solidDryer</code>	Gas–solid drying (constant + falling rate)	T_{gas} , sample drying curve
<code>crystalliser</code>	Population-balance crystalliser	nucleation/growth kinetics
<code>cyclone / bagFilter / gasSolidSplitter</code>	Gas–solid separation (5 cyclone sub-models)	geometry per model
<code>spiralWoundModule</code> (<i>alias</i> <code>membraneSW</code>)	NF/RO spiral-wound element (solution-diffusion)	membrane name, area, P_{permeate} , k_{film}

Each unit also accepts (top-level, not inside `operation`) overrides for solver tolerances — same keys as in `solverDict`.

Energy ports and heat integration. Rotating equipment, heaters, and distillation reboilers/condensers exchange *work* and *heat* through typed energy ports (`energyOutputs` / `energyInputs`) rather than process streams. A turbine's shaft work can drive a compressor or an `electricLoad`; a column's reboiler heat can preheat another unit (a forward heat-link). This layer — the W/Q wires, the `column.condenser` / `column.reboiler` port names, the auto-allocation-by-temperature rule, and the `utilityAllocation` report — is documented in `docs/ai/energy.md`.

5 Built-in thermodynamic models

What you'll learn. The four families of models that together cover everything the simulator does: activity coefficients, equation of state, vapour pressure, and heat capacity.

5.1 Activity models (liquid γ)

model	Notes
ideal	$\gamma_i = 1$ — the Raoult limit.
NRTL	Renon-Prausnitz; needs binary pairs.
Wilson	Wilson 1964; needs binary pairs + V_{liq} per component.

NRTL pair entry:

```

pairs
(
  {
    i      ethanol;      j      water;
    a_ij   -0.8009;      b_ij   246.18;      // tau_ij = a_ij + b_ij/T
    a_ji   3.4578;      b_ji   -586.0809;
    alpha  0.30;
  }
);

```

5.2 Equation of state (vapour φ)

model	Notes
idealGas	$\varphi_i = 1$; $PV = RT$. Adequate for the low-pressure tutorials.

5.3 Vapour pressure (per component)

Stored in `data/standards/components/<name>.dat`:

model	Form
Antoine	$\log_{10} P^{\text{sat}}[\text{bar}] = A - B/(T[\text{K}] + C)$

5.4 Heat capacity (per component, optional)

model	Form
polynomial	$C_p = a_0 + a_1T + a_2T^2 + a_3T^3 + a_4T^4$, [J/(mol·K)]

5.5 Phase types (multi-phase / LLE)

type	Notes
vapor	wraps an EoS
liquid	wraps an ActivityModel + uses Antoine for P^{sat}
solid	stub: declared interface for future SLE / inert solids

6 Sensitivity, optimisation, parameter estimation

What you'll learn. How to wrap the simulator in an outer loop without writing a single line of C++.

Drop a `system/outerDict` into a case and the simulator is wrapped in an outer loop.

6.1 Sensitivity sweep — type sweep;

Vary one design value across a grid and harvest responses:

```
type          sweep;

parameter
{
    target     units[0].operation.refluxRatio;    // JSON-path-like syntax
    range      ( 1.5  5.0 );
    nPoints     10;
}

responses
(column01.T_top
 column01.T_bottom
 column01.x_D_LK
 column01.V_strip
);

report
{
    format     csv;
    file        sweep_results.csv;
}
```

Each response is `unitName.kpiName` (a unit-op KPI) or `streamName.field` where field is F, T, P, or vf.

Try it now. `runCase tutorials/steady/optimisation/sensitivity01_column_reflux` produces `sweep_results.csv`. Plot column 6 against column 2 with any tool; you'll see the classic " V_{strip} asymptote" as the reflux ratio grows past the minimum.

6.2 Parameter estimation

A dedicated concrete driver, `type fitBinaryPair;`, regresses the four interaction parameters of one binary pair against experimental bubble-temperature data using hand-rolled Levenberg-Marquardt (Marquardt diagonal damping, central finite-difference Jacobian, 3-up/5-down trust-region update of λ):

```
type          fitBinaryPair;

pair          ethanol-water;
data          constant/experiments/ethanol-water-101kPa;
unit          bt01;          % the bubbleT unit that emits T_bubble

% Optional (defaults shown):
% maxIter     30;           tolerance 1.0e-6;
% fdStep      1.0e-4; lambda0    1.0e-3;
```

The dataset is a dict with a flat two-column list:

```
P      1.01325;
data
(
    0.05  361.95
    0.10  359.95
    ...
    0.95  351.45
);
```

The driver writes a dated proposal to `constant/binaryPairs/<model>/<c1>-<c2>.fit-<YYYY-MM-DD>.dat`; the human promotes it manually with `mv` once inspected. Worked example: `tutorials/fitNRTL01_ethanol_water/`, with detailed walk-through in its `README.md`.

6.3 Optimisation — type optimization;

`type optimization`; drives a hand-rolled Nelder-Mead simplex to minimise a target KPI (e.g. annualised cost) over one or more decision variables, re-running the whole simulator — and, if a `postDict` is present, the sizing + costing chain — at every evaluation. Two worked examples ship: `tutorials/steady/optimisation/optim01_column_reflux` (minimises `column01.V_strip` over reflux ratio + feed stage) and `tutorials/steady/optimisation/optim02_process_cost` (minimises the total Guthrie module cost over reactor volume + heater outlet temperature).

7 Equipment sizing and costing

What you'll learn. How a single `postDict` turns a converged flowsheet into a vessel list, wall thicknesses, weights, and Guthrie/Turton bare-module costs in EUR.

Drop a `system/postDict` and the simulator result is post-processed.

7.1 Sizing — `sizing { units (...) }`

```
sizing
{
  units
  (
    {
      unitName    reactor;
      type        stirredTank;
      material     SS316;
      designRules
      {
        L_over_D      2.5;
        pressureDesign 3.0;
        corrosionAllow 0.003;      // m
        jointEfficiency 1.0;
      }
    }

    {
      unitName    heater;
      type        shellTubeHX;
      material     SS304;
      designRules
      {
        U            600.0;      // W/(m^2 K)
        LMTD         10.0;      // K
        pressureDesign 2.0;
      }
    }
  );
}
```

`type` selects an `EquipmentSize` model. At: `stirredTank` (vessels including CSTRs and reactor surge drums) and `shellTubeHX`. Materials come from `data/standards/materials/*.dat` — built-in: `carbonSteel`, `SS304`, `SS316`, `aluminium`.

7.2 Costing — `costing { method... }`

```
costing
{
  method      Guthrie;
  year        2026;
  cepci       820;
  cepci2001   397;
  usdToEur    0.92;
}
```

The `Guthrie` model uses Turton-style purchased-cost correlations, ASME wall-thickness pressure factor for vessels, polynomial F_P for heat exchangers, then $C_{BM} = C_p(B_1 + B_2 F_M F_P)$, and $C_{TM} = 1.18 C_{BM}$.

The post-processor prints a per-unit cost table to stdout:

```
===== Equipment Costing =====
Method: Guthrie   Year: 2026.0   CEPCI: 820.0
```

unit	equipment	F_M	F_P	C_purchased	C_bare_mod	C_total_mod
heater	shellTubeHX	2.34	1.09	365 249	2 146 753	2 533 169
reactor	stirredTank	3.05	1.00	2 055	16 033	18 918
TOTALS (EUR)				367 304	2 162 786	2 552 087

8 Tour of the tutorials

What you'll learn. Choupo ships 131 tutorials under `tutorials/{steady,batch,ctrl,props,plant}/`; `listCases` tabulates them all with one-line descriptions. The selection below is the steady-state on-ramp — pick the one closest to what you want to do and edit from there.

Tutorial	What you learn
<code>flash01_benzene_toluene</code>	First run: Raoult VLE, Rachford-Rice, Newton iteration visible.
<code>flash02_ethanol_water</code>	NRTL + Wegstein outer loop (~ 7 iter vs ~ 29 direct subst.).
<code>flash03_wilson_ethanol_water</code>	Same problem with Wilson — model uncertainty visible.
<code>bubbleT01_ethanol_water</code>	Find T such that $\sum K_i x_i = 1$.
<code>dewT01_ethanol_water</code>	Find T such that $\sum y_i / K_i = 1$.
<code>adiabaticFlash01_benzene_toluene</code>	$Q = 0$ — solve for T_{out} ; uses C_p and ΔH_{vap} (Watson).
<code>cstr01_first_order</code>	Newton on extent ξ ; Damköhler & analytic check.
<code>pfr01_first_order</code>	RK4 by hand; reproduces $X = 1 - e^{-k\tau}$.
<code>hx01_heater</code>	Specify T_{out} or Q , compute the other; Newton 1-D.
<code>process01_reactor_flash</code>	First flowsheet: CSTR $\rightarrow$ flash; mass-balance closure.
<code>process02_reactor_hx_flash</code>	Three units; T propagates through the chain.
<code>process02_with_design</code>	The above + sizing + Guthrie costing — full CAPEX in one run.
<code>process03_recycle</code>	Reactor-separator-recycle: tear stream + Wegstein outer loop.
<code>column01_benzene_toluene</code>	Multi-stage column, Wang-Henke bubble-point.
<code>sensitivity01_column_reflux</code>	Sweep R over $[1.5, 5]$ to CSV with V_{reboiler} vs purity.
<code>optim01_column_reflux</code>	Nelder-Mead: minimise V_{strip} over reflux + feed stage.
<code>optim02_process_cost</code>	Nelder-Mead on total Guthrie module cost (sizing + costing per evaluation).
<code>process04_research_workflow</code>	Case-local <code>constant/binaryPairs/<model>/<pair>.dat</code> instead of inline pairs — the lookup-order pattern.
<code>fitNRTL01_ethanol_water</code>	<code>fitBinaryPair</code> outer driver: LM regression of NRTL ethanol/water against 1 atm bubble- T data (see §6.2).
<code>extraction01_waterButanol</code>	LL extraction with NRTL — <i>experimental</i> (see §12).

9 Output anatomy

What you'll learn. What every section of a run's log means, and why the inner iteration trace is the part you should read.

A typical run with `verbosity 3` prints, in order:

1. **Banner** — version + licence.
2. **Case header** — case dir, database root, optional-dicts presence.
3. **Components and phases** declared.
4. **Topology** of the flowsheet — source streams + the per-unit blocks.
5. **Per-unit run** — for each unit operation:
 - Inputs echo ($F, T, P, \mathbf{z}$).
 - Inner iteration trace (Newton steps with $g(x)$, dg/dx , ΔV).
 - For γ -models: outer Wegstein loop with γ values per iter.
 - Result block (regime, V/F , $\mathbf{x}$, $\mathbf{y}$, $\mathbf{K}$).
6. **Flowsheet summary** — final stream table.
7. **Outer driver / post-processor output** (if applicable).

Key idea. The inner iteration trace is where the simulator stops being a black box. Convergence ratios in the trace tell you whether Newton is converging quadratically (good) or linearly (problem is ill-conditioned).

10 The web GUI: a runner and visualiser

What you'll learn. How to open a case in the browser, run it, and read the result — and why the GUI never edits your dicts.

Choupo ships a browser front-end under `gui/`. Its design stance is deliberate and worth stating up front: **the GUI is a runner and visualiser, not an editor**. Cases are authored as plain-text dicts under `tutorials/<name>/system/` and `constant/`, edited with any text editor (increasingly with LLM help). The GUI loads those dicts, draws the flowsheet, runs the solver *in the browser*, and renders the log + plots. The dicts on disk remain the single source of truth; per-case scalar tweaks in the GUI are downloaded back to disk, never silently mutated. The architectural inspiration is ParaView (a DSL plus a viewer), not a block-diagram GUI editor.

10.1 Starting the GUI

```
make wasm-gui      # build the WASM solver (choupoSolve + choupoProps)
bin/runGui         # boots the dev server and opens a browser window
bin/runGui --kill  # stops everything
```

10.2 The layout

The shell is a thin top bar over a full-screen flowsheet canvas — no left tree, no docked right panel, no bottom tab strip. Auxiliary views (log, plots, stream table) open on demand as workspaces. Selecting a unit or stream raises a floating selection card in the top-right corner; the card is read-only (labels, units, help tooltips driven by the operation schemas).

10.3 Interactions

- **Single-click** a node or edge — raises the floating selection card.
- **Double-click** — pops the unit out (or drills into a composite sector).
- **Drag** nodes, edge bends, and connection points to lay the flowsheet out; the layout persists per case in the browser's `localStorage`.
- **Stream-class toggle** — show/hide whole classes of streams (e.g. energy wires) to declutter the canvas.
- **Run** — streams the same log you would see from `runCase`; **Stop** cancels mid-solve.

Stream colour encodes phase and comes straight from the solver (`vf` + solid mass fraction); it never changes on hover or selection — selection is signalled by a halo/ring, not a recolour.

Common pitfall. MOCK vs WASM. If the WASM solver has not been built (`make wasm-gui`), the GUI silently falls back to a *mock* adapter that returns canned numbers so the UI still renders. The Run badge reads **MOCK** in that case, not **WASM** — check it before trusting any result. After changing a unit op or any C++ source, rebuild the WASM (it is a separate artefact from the native binary) or the browser keeps running stale code.

11 Building your own case

What you'll learn. The fastest path from a blank idea to a converged result is to clone the tutorial closest to your idea and edit.

```
cp -r tutorials/steady/flash/flash01_benzene_toluene tutorials/steady/myFlash
$EDITOR tutorials/steady/myFlash/system/flowsheetDict
$EDITOR tutorials/steady/myFlash/constant/thermoPackage
runCase tutorials/steady/myFlash
```

Adding a missing component. Two locations are available — pick the right tier for what you are adding:

- `data/standards/components/<name>.dat` — contribution to the *shared catalogue* of well-curated chemicals.
 - `<case>/constant/components/<name>.dat` — *sample-specific or one-off data*: a particular powder's sorption isotherm, a custom membrane solute, a laboratory measurement. Per-case files are overlaid field-by-field on top of the standard catalogue, so a slim file with just the bespoke fields is enough — the rest is inherited from the standard entry of the same name.
1. Create the file in the chosen tier, following the same format as the existing standard entries (see Developer Guide § Adding components for the field list). When the file carries a `gibbsFormation` block, declare the **phase** of the tabulated Δh_f° explicitly (`gas` / `liquid` / `solid`) — the value tells the solver which rung of the enthalpy ladder the data is pinned to; see Theory Guide § “Which rung does the data file pin?”.
 2. Add `<name>` to the `components (...)` list in your `thermoPackage`.

Using a plant utility. Heating and cooling services are catalogued separately from process components, under `data/standards/utilities/`. Eight services ship by default — LP / MP / HP steam, cooling water, chilled water, Dowtherm A, HITEC molten salt, and a propylene-glycol refrigeration brine — each with operating conditions and a `cost` in EUR/GJ.

A stream block pulls the catalogue defaults with a single keyword:

```
streams
{
    PlantSteam { utility steamLP; F 60 kmol/h; }
}
```

The parser substitutes `components`, `state`, `P` and `T` from the `steamLP.dat` entry (the user keeps full control — writing `P 3 bar`; alongside the `utility` key overrides the catalogue default for that one case), and marks the stream's internal category to `"steamLP"`. When the case runs, the `utilities` report aggregates every utility-tagged stream by category and emits `reports/utilities/consumption.csv` with the kg/h, MW, and EUR/h delivered by each header. Multiple streams sharing a catalogue name (two heaters tapping the same plant header) sum automatically.

Adding a missing material. Same procedure under `data/standards/materials/`.

Using a reaction in a CSTR / PFR. Declare it in `constant/reactions` and reference it by name in the unit's `reaction <name>;` entry.

12 Troubleshooting

What you'll learn. The seven failure modes that account for nearly every support ticket, and exactly what to do about each.

“Component does not exist” / “Material does not exist”

```
ERROR: Database: component file not found: data/standards/components/foo.dat
```

The component (or material) name is misspelt, or the file is missing. Filenames are case-sensitive. Check `data/standards/components/`.

Build fails after pulling new code

```
make distclean
make all
```

Removes the entire `build/` tree. Necessary when headers change in incompatible ways.

Simulator runs but flowsheet returns NaN

Most common: feed composition does not sum to 1, or a unit-op is fed with an empty stream. Re-read the input echo and verify $F, T, P, \mathbf{z}$ values.

Newton fails to converge

Two situations:

- **Outer Wegstein** (γ -model loop): `qmin` too aggressive. Try `outerAccelerator directSubstitution` for non-ideal LL or azeotropic systems.
- **Inner Newton** (Rachford-Rice or distillation bubble-T): increase `maxIter` and `maxOuterIter`. Near the phase boundary, tighten `compositionTol`.

LL extraction tutorial does not give a physical split

Known issue (still open at) — LL flash converges to the trivial $K = 1$ fixed point in systems with the same γ -model on both phases. Resolution requires second-order Michelsen TPD with Newton — scheduled. The tutorial is shipped as *experimental* and documents this gap.

Sweep run fails on some points

Almost always: the simulator's `maxOuterIter` is too small for the edge points (high reflux ratio, high pressure, ...). Increase `maxOuterIter` in the corresponding unit-operation block in `flowsheetDict`.

Common pitfall. A run that prints $V/F = \text{nan}$ mid-flash almost always means $\log P_i^{\text{sat}}$ was evaluated outside Antoine's valid range (e.g. T too low so Antoine returns a negative pressure), or that a stream upstream had $F = 0$ and emptied the composition vector.

13 Where to look next

- `CLAUDE.md` — conventions, architecture overview, AI-assistant guide.
- `docs/developerGuide.pdf` — adding unit operations, thermo models, components, materials, outer drivers, post-processors.
- `tutorials/<name>/system/*` — every tutorial is a fully-worked example.
- `src/applications/choupoSolve/main.cpp` — the entry point.

Key idea. If a case behaves unexpectedly, *read the log*. Every Newton step, every Wegstein update, every γ value is printed. Black-box debugging is exactly the failure mode Choupo refuses to participate in.

Colophon. This manual was typeset in Latin Modern with the Choupo shared LaTeX preamble. Sources live in `docs/`; rebuild with `cd docs && make`.